

Toxicity of 6PPD alternatives to salmonid cell lines

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Introduction

Stormwater runoff in urban areas introduces numerous anthropogenic chemicals into surrounding aquatic environments. One such chemical is 6PPD (*N*-(1,3-Dimethylbutyl)-*N'*-phenyl-*p*-phenylenediamine), the primary antidegradant responsible for protecting tire rubber from ozone-induced degradation and cracking. When exposed to ozone on the road surface 6PPD is transformed into the toxic transformation product 6PPD-quinone (6PPDQ). This compound is highly lethal to coho salmon (*Oncorhynchus kisutch*), and 24 hour 50% lethal concentrations (LC₅₀; 41-95 ng/L) are frequently detected in urban streams during stormwater runoff events¹⁻⁵. The highest stream concentrations of 6PPDQ and most extensive coho salmon mortality occur in high-traffic urban areas, where tire wear particles accumulate due to the frequent braking and acceleration of cars⁶. In the Seattle area, some coho salmon-bearing streams can lose over 90% of returning females prior to spawning due to 6PPDQ-containing runoff⁷. Prespawn loss of this magnitude could result in the extinction of some urban spawning populations⁸. Importantly, the lethal effects of 6PPDQ have been observed in several other salmonid species, indicating that its impact may threaten fish health for urban aquatic environments globally⁹⁻¹¹.

The ubiquity of 6PPD in automobile tires, high toxicity of the transformation product 6PPDQ, and prevalence of 6PPD and 6PPDQ in environmental matrices has spurred investigations into alternative rubber antiozonants to replace 6PPD that retain tire performance while reducing ecological harm. A technical memorandum provided to the legislature by the Washington State Department of Ecology summarized a list of potential 6PPD alternatives requiring further study, including several structurally similar *p*-phenylenediamines (PPDs)¹². However, the structural similarity of other PPDs to 6PPD and their known production of quinone transformation products¹³ raises concerns that they may elicit similar toxic effects as 6PPDQ. Currently, investigations into the toxicity of proposed PPD alternatives or their ozonated transformation products in coho salmon are limited, representing a significant data gap in evaluating whether they offer improved environmental safety over 6PPD.

This project investigated the toxicity of proposed alternative rubber antiozonants *in vitro* using immortalized cell lines derived from three salmonid species with known differences in sensitivity to 6PPDQ (coho salmon, Chinook salmon (*O. tshawytscha*), rainbow trout (*O. mykiss*))¹. These

immortalized cell lines replicate toxic responses observed in *in vivo* studies, while enabling higher-throughput testing and reducing the need for animal use and other resource-intensive procedures. The antiozonants and their transformation products selected for testing were chosen based on multiple criteria, including their inclusion in the Ecology technical memorandum, structural substitutions (e.g., branched/cyclic alkyl vs aryl substitutions), commercial availability, and input from the Washington State Department of Ecology and other industry experts.

Materials and Methods

Stock chemicals

CCPDQ was received as a donation from Lanxess AG (Cologne, Germany). 7PPD and 77PD were received as a donation from Flexsys (Akron, OH). All other chemicals were purchased from HPC Standards (Atlanta, GA) or Cayman Chemical (Ann Arbor, MI) as indicated in **Table 1**. Stock solutions of parental antiozonants were dissolved overnight in absolute ethanol on a shaker immediately prior to testing. Ethanol was chosen as a solvent for parental antiozonants to match the solvent used for *in vivo* antiozonant testing being performed under IAA No. C2100043. For quinone transformation products, dimethyl sulfoxide (DMSO) was used as the solvent for direct comparison with previously published studies on 6PPDQ toxicity in salmonid cell lines.

Table 1. Alternative antioxidants and quinone transformation products assessed for *in vitro* toxicity to salmonids. Lot numbers and purity presented are for the production batch used for toxicity testing.

<u>Parent antiozonants</u>	<u>CASRN</u>	<u>Source</u>	<u>Lot #</u>	<u>Purity*</u>
6PPD [N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine]	793-24-8	Cayman	N/A	99.1%
7PPD [N1-(1,4-dimethylpentyl)-N4-phenyl-1,4-benzenediamine]	3081-01-4	Flexsys	N/A	97.0%
77PD [N1,N4-bis(1,4-dimethylpentyl)-1,4-benzenediamine]	3081-14-9	Flexsys	N/A	98.4%
IPPD [N-Isopropyl-N'-phenyl-1,4-phenylenediamine]	101-72-4	HPC	822474	98.01%
CCPD [N,N'-Dicyclohexyl-4-phenylenediamine]	4175-38-6	Flexsys	N/A	N/A
DPPD [N,N'-Diphenyl-p-phenylenediamine]	74-31-7	Cayman	0704663	99.07%

<u>Quinone products</u>				
6PPDQ [N-(1,3-dimethylbutyl)-N'-phenyl-p-phenylenediamine-quinone]	2754428-18-5	HPC	818631	98.65%
7PPDQ [2-[(1,4-dimethylpentyl)amino]-5-(phenylamino)-2,5-cyclohexadiene-1,4-dione]	2894124-00-4	Cayman	0713246	>95%
77PPDQ [N,N'-bis(1,4-dimethylpentyl)-p-Phenylenediamine-quinone]	N/A	HPC	813463	96.74%
IPPDQ [N-Isopropyl-N'-phenyl-p-phenylenediamine-quinone]	98054-73-9	HPC	826491	98.11%
CCPDQ [2,5-bis(cyclohexylamino)-1,4-benzoquinone]	1653-85-6	Lanxess	N/A	>90%
DPPDQ [N,N'-Diphenyl-p-phenylenediamine-quinone]	3421-08-7	Cayman	0705217	>95%

*Reported in Certificate of Analysis supplied by manufacturer; not independently verified
N/A indicates information that was unavailable

Gas-phase ozonation

Gas-phase ozonation was performed on two PPDs (7PPD, IPPD) to generate transformation product mixtures. 6PPD was ozonated as a positive control to provide a basis for comparing the toxicity of 7PPD and IPPD to their ozonated mixtures. Before ozonation, parent compounds were dissolved in methanol and applied to 8" x 8" glass plates using a gas-tight syringe at 100 µg PPD/cm². After the chemicals were applied, the plates were allowed to dry under a fume hood until the methanol was evaporated. Once dry, the plates were placed in a glass chamber receiving 0.5 L/min of positive pressure compressed air containing 50 ppmv ozone. Ozone concentrations entering and exiting the glass chamber were continuously monitored using a UV/Vis spectrophotometer.

Ozonation was considered complete after ozone consumption by the PPD mass on the plates had effectively ceased. Following complete ozonation, the plates were removed from the chamber and eluted using absolute ethanol. The eluent was transferred into pre-weighed amber glass bottles and purged with N₂ before storage at -20°C. For each ozonation experiment, one control plate with the same mass of PPD applied was placed in a vacuum chamber with compressed air for the duration of the ozonation process and eluted at the end of the experiment. The control and

ozonated eluents were analyzed using targeted mass spectrometry to quantify the masses of the unreacted parent compounds and quinone transformation product. The ozonation of IPPD was performed twice due to rapid reaction rates. During the first ozonation, ozone reaction ceased at 5 hours, but the plate was ozonated for 24 hours to align with the ozonation times of 6PPD and 7PPD. Higher purity IPPD (TCI, 97.2% purity) was used for the second ozonation. The reaction again ceased early (6 hours), and the experiment was stopped.

Cell line exposures

The alamarBlue cell viability kit (Invitrogen) was used to quantify cellular metabolism as previously described¹. Briefly, immortalized cell lines derived from coho salmon (CSE-119), Chinook salmon (CHSE-214), and rainbow trout (RTG-2) were seeded in 96-well plates at a density of 10,000 cells per well in 100 μ L of L15 media supplemented with 10% fetal bovine serum (Peak Serum Inc.) and 10% penicillin-streptomycin (Cytiva HyClone) and allowed to adhere for 24 hours. Stock solutions for parental antiozonants and purified quinone transformation product were serial diluted to 2X the desired final concentrations and added in 100 μ L increments to each well to create nominal exposure concentrations, with a minimum of 8 concentrations tested per cell line. Transformation product mixtures for 7PPD, IPPD, and 6PPD were applied up to a maximum concentration of 200 μ g/L of the parent PPD.

Cell media containing the same percentage of ethanol or DMSO as the highest chemical concentration (0.0067% DMSO or EtOH) served as a solvent control. Each plate contained 4 replicates per concentration per cell type. Following chemical addition, the alamarBlue cell viability reagent was added to each well at 10% of the well volume in a dark room per the manufacture's protocol. Fluorescence was measured at an excitation of 550 nm and emission of 600 nm with a gain of 70 using the Tecan Magellan Pro Infinite M Plex plate reader at the 0-, 6-, and 24-hour timepoints (+10 min). Plates were kept covered with foil in a 20°C incubator between measurements. Relative metabolism compared to solvent controls was calculated using measured fluorescence at 24 h compared to solvent controls from the same experiment at 6 h post-exposure.

Quality Assurance / Quality Control

Metabolic activity in unexposed cells was assessed in quadruplicate on every plate for each cell line tested. To ensure unexposed cells were healthy and performing the expected metabolism of alamarBlue, a minimum of a two-fold increase in fluorescence was required at 24 h when compared to the same wells immediately after addition of the reagent. To ensure that the solvent (DMSO or absolute ethanol) did not induce toxicity at the highest concentrations tested, solvent controls were compared to unexposed cells. A successful experiment was defined when the solvent induced < 15% reduction in metabolism compared to unexposed controls. 6PPDQ exposures were also performed intermittently to confirm cell health and expected response based on previous studies.

Results

Parent Antiozonants

The parental PPDs examined did not elicit significant toxicity in the tested salmonid cell lines (Fig. 1).

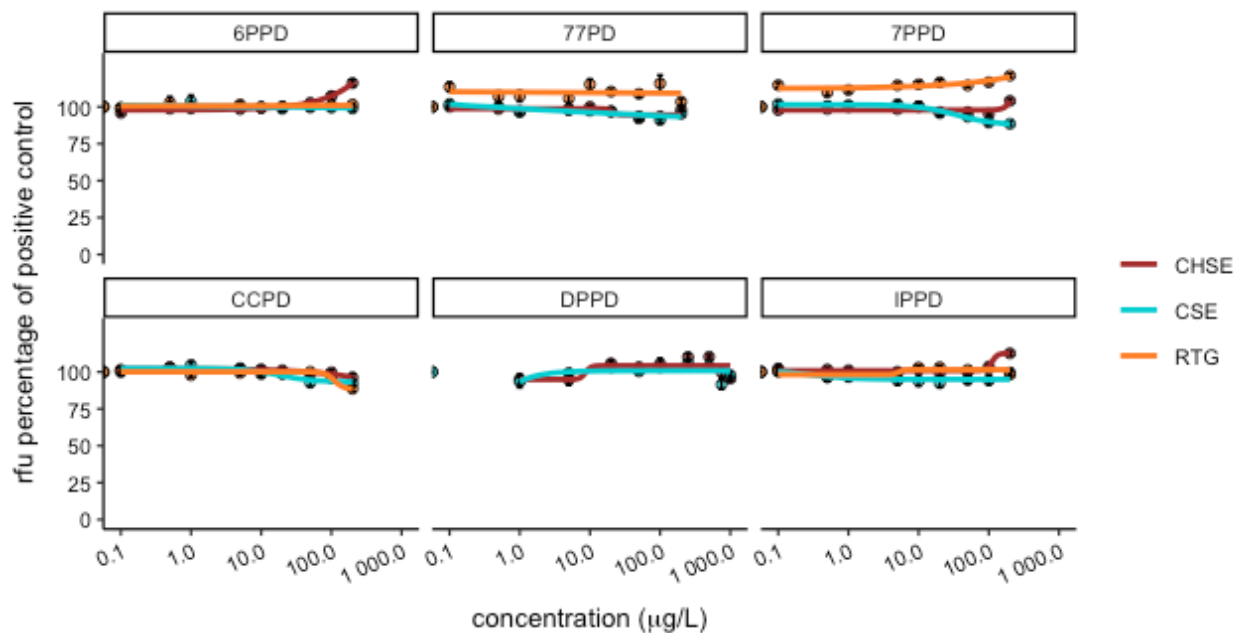


Fig. 1. Metabolic response of salmonid cell lines to parental antiozonants. Each point corresponds to the mean change in metabolism compared to solvent exposed control cells. Error

bars represent $\pm$ standard error of the mean (SEM). Standard error was typically smaller than the size of the points and are therefore not visible in the figure. X-axis is displayed on a $\log_{10}$ scale. CHSE – Chinook salmon, CSE – coho salmon, RTG – rainbow trout

Quinone Products

As expected, 6PPDQ was highly toxic to coho salmon CSE-119 cells (50% effect concentration (EC_{50}) = 9.97 $\mu\text{g/L}$) but not CHSE-214 cells (**Fig. 2**)¹. RTG-2 cells did not show metabolic disruption with 6PPDQ exposure up to 100 $\mu\text{g/L}$. This is in slight contrast to our previous findings, where concentrations > 80 $\mu\text{g/L}$ induced $\sim 10\%$ reductions in metabolic activity¹. In addition to 6PPDQ, reductions in metabolic activity were observed in CSE-119 cells for 7PPDQ and IPPDQ at higher concentrations (**Fig. 2, Table 2**). Both compounds elicited reductions in metabolic activity greater than 20% at concentrations below 500 $\mu\text{g/L}$. None of the six quinones tested had adverse effects on metabolic function in CHSE-214 or RTG-2 cells (**Fig. 2**).

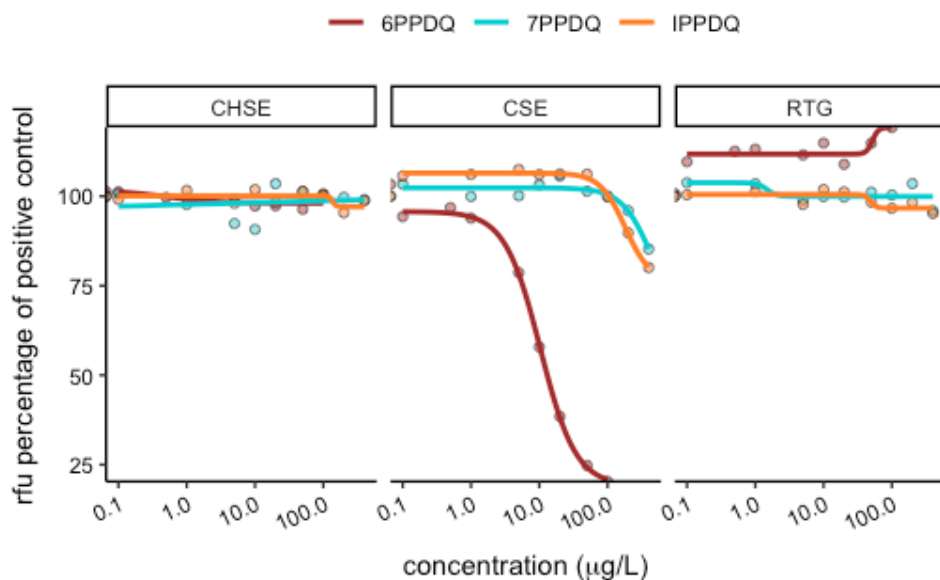


Fig. 2. Metabolic effects of purified PPD quinones on salmonid cell lines. DPPDQ, 77PDQ, and CCPDQ did not elicit observable effects and are excluded from the figure for clarity. X-axis is displayed on a $\log_{10}$ scale. CHSE – Chinook salmon, CSE – coho salmon, RTG – rainbow trout

Table 2. Quinone nominal concentrations eliciting 5, 10, and 20% effects (EC₅, EC₁₀, EC₂₀) on metabolic function in coho salmon CSE-119 cells. Concentrations are presented in µg/L.

<u>Cell type</u>	<u>Chemical</u>	<u>EC₅</u>	<u>EC₁₀</u>	<u>EC₂₀</u>
CSE	6PPDQ	1.33	2.38	4.38
CSE	7PPDQ	223.38	315.83	490.73
CSE	IPPDQ	147.85	194.73	408.01

Transformation Product Mixtures

Mixtures of ozonated PPD transformation products were generated for 6PPD, 7PPD, and IPPD to assess the impacts of ozonated mixtures that contain quinone and other ozonated PPD products. During ozonation, the conversion of IPPD to IPPDQ was substantially lower than for 6PPD or 7PPD, resulting in lower maximum concentrations that could be tested (**Table 3**).

Regardless, in CSE-119 cells all three ozonated mixtures elicited at least a 20% reduction in metabolism (EC₂₀) at the maximum concentrations tested (**Figs. 3, 4, Table 3**). At least 20% reductions in metabolism were also observed in CHSE-214 cells for 7PPD, and for both 7PPD and IPPD in RTG-2 cells.

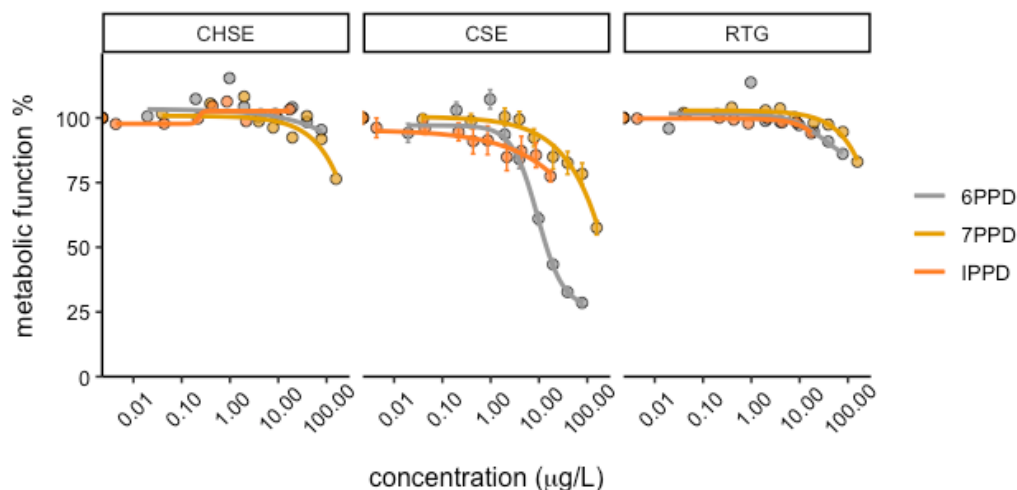


Fig. 3. Metabolic effects of ozonated PPD mixtures on salmonid cell lines. Concentrations displayed on x-axis are the concentration of the quinone transformation product, measured by

analytical chemistry after ozonation but prior to toxicological testing. Quinone concentrations displayed are therefore nominal concentrations.

CHSE – Chinook salmon, CSE – coho salmon, RTG – rainbow trout

Table 3. Maximum concentrations of quinone that were able to be tested in the ozonated mixtures without inhibiting basal metabolic function.

<u>Chemical</u>	<u>Maximum quinone conc. (µg/L)</u>
6PPDQ	95.4
7PPDQ	156.8
IPPDQ	17.3

* All values derived are adapted from Lawrence 2025¹⁴

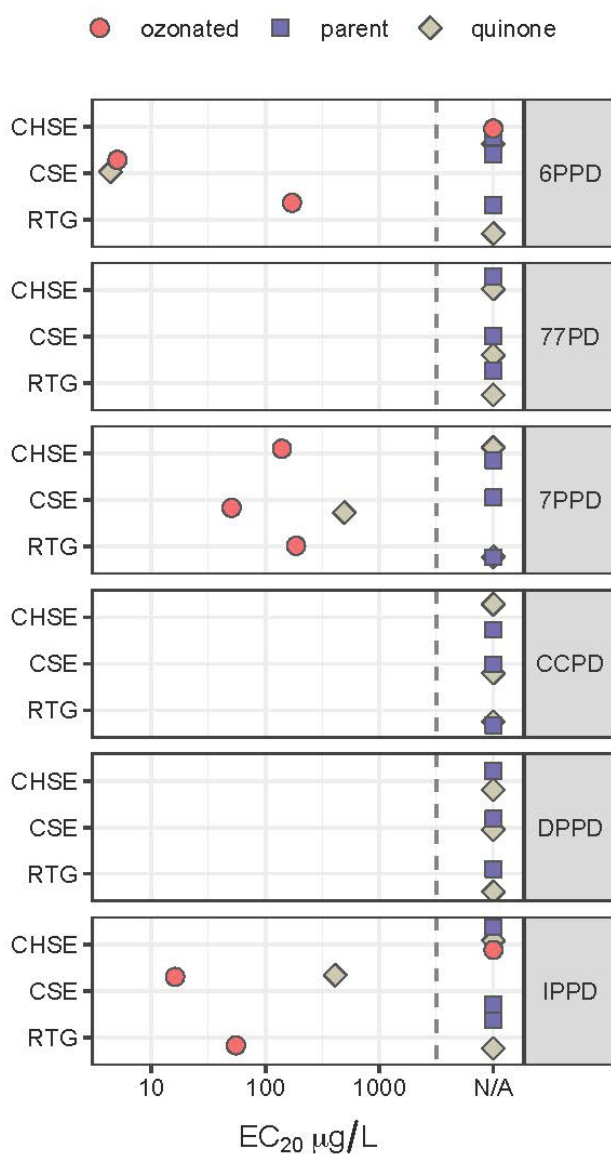


Fig. 4. Summary of EC₂₀ concentrations for each chemical tested. Concentrations presented are for the parental compound for parent chemical and quinone concentrations for quinone and ozonated transformation mixtures. X-axis is log10 scale. N/A indicates no alterations in metabolic function up to the maximum concentration tested, and therefore no EC₂₀ was calculated.

CHSE – Chinook salmon, CSE – coho salmon, RTG – rainbow trout

Conclusions

The acute toxicity of 6PPDQ to coho salmon and the high sensitivity of other salmonid species have prompted research into alternative tire antiozonants that can provide similar tire performance while mitigating ecological impact. Although a range of chemical classes of antidegradants have been proposed, other PPDs are currently regarded as the most promising candidates for maintaining the functional requirements of tires. Here, none of the parental PPDs tested induced measurable toxicity in any cell line. This corroborates previous studies suggesting that parental PPDs are themselves unlikely to cause toxicity or alter metabolic activity.

Water chemistry was not measured during exposures, which could be important for parent PPD chemicals specifically as they are highly reactive and may be unstable in aqueous solutions. Concentrations of parental PPDs therefore likely declined substantially over the exposure period. However, this is expected to mimic the *in situ* behavior of these chemicals in watersheds and are therefore relevant to real world exposures.

The toxicity to coho salmon cells induced by 6PPDQ in this study occurred at comparable concentrations ($EC_{50} = 9.97 \mu\text{g/L}$) to previous studies¹, and all parental PPDs, quinone products, and ozonated mixtures tested were found to be less toxic than 6PPDQ. However, in addition to 6PPDQ, both 7PPDQ and IPPDQ elicited at least a 20% reduction in metabolism in coho salmon cells. Given their structural similarity to 6PPDQ and toxicity exclusive to coho salmon cells, the biological targets responsible for the toxic effects of all three PPD quinones are likely shared. This hypothesis is further supported by a relative lack of biological activity for IPPDQ and 7PPDQ in Chinook salmon and rainbow trout cells, consistent with their presumed lack of the biological target for 6PPDQ. Immortalized coho salmon cells are known to be ~70-fold less sensitive than *in vivo* studies, and thus toxicity observed here warrants further investigation in sensitive species.

In situ ozonation of PPDs produces numerous transformation products¹⁵, many of which have unresolved chemical structures and unknown biological activity. We observed that ozonated mixtures of 6PPD, 7PPD, and IPPD were typically more toxic than the purified quinones (**Fig. 4**), which implies adverse effects mediated by other PPD transformation products. For example, ozonated 7PPD altered metabolic activity in Chinook salmon cells despite being unresponsive to

purified 7PPD or 7PPDQ. All three ozonated mixtures also showed greater toxic effects in RTG-2 cells compared to purified quinones. However, it is important to note that non-target chemical analysis was not performed on ozonated mixtures produced in this study, and therefore the toxic constituent(s) in each mixture are unclear. It is possible that other contaminants were present or ozonation produced other products not yet identified. Additional work would be necessary to elucidate the drivers of toxicity in these mixtures.

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